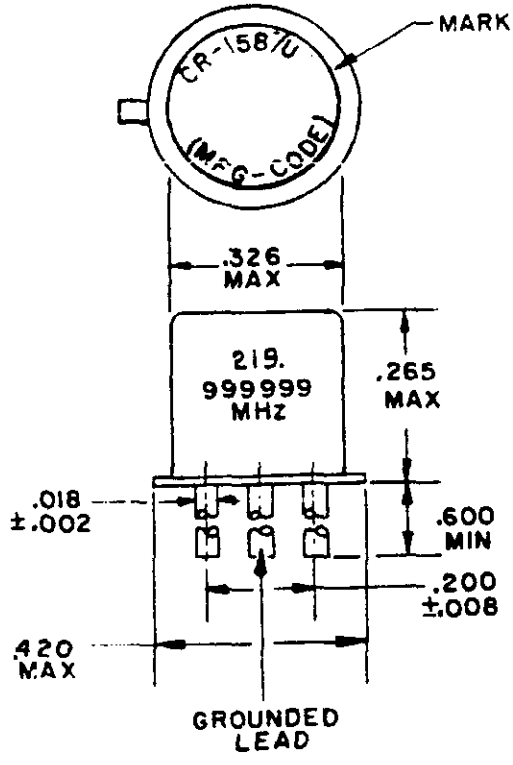
MILITARY SPECIFICATION SHEET

CRYSTAL UNITS, QUARTZ, CR-158/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 200 to 220 MHz; seventh overtone; noncontrolled; series.



INCH	MM
.002	.05
.008	.20
.018	.46
.200	5.08
.265	6.73
.326	8.28
.420	10.67
.600	15.24

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The crystal unit shall not be trimmed after sealing.

FIGURE 1. Crystal unit - CR-158/U.

(B) denotes changes

FSC 5955

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TABLE I. Requirements.

Requirements	Specified value or condition
Holder	HC-35/U, MIL-H-10056/21
Dimensions, marking, and configuration	See figure 1
Frequency range	200 to 220 MHz, inclusive
Frequency tolerances:	
Operating temperature range	±50 ppm
Room temperature	±30 ppm
Equivalent resistance	150 ohms, maximum
Mode of oscillation	Seventh mechanical overtone
Temperature ranges:	
Operating	+80°C to +90°C, inclusive
Operable	-55°C to +105°C, inclusive
Test set, calibration values, and rated drive level	See table II
Capacitance, shunt	4.5 pF, maximum
Shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Thermal shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	±5 ppm

1/ Method 204, MIL-STD-202, test condition A.

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	MHz	Ohms	mA	Volt	mW
AN/TSM-15	200 to 220	100	--	0.1	0.1

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(B) QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

(B) Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5955-

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